

A Security Study about Electron Applications and a Programming Methodology to Tame DOM Functionalities

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The Electron platform represents a paradigm to develop modern desktop apps using HTML and JavaScript. Microsoft Teams, Visual Studio Code and other flagship products are examples of Electron apps. This new paradigm inherits the security challenges in web programming into the desktop-app realm, thus opens a new way for local-machine exploitation. We conducted a security study about real-world Electron apps, and discovered many vulnerabilities that are now confirmed by the app vendors. The conventional wisdom is to view these bugs as *sanitization errors*.

Accordingly, secure programming requires programmers to explicitly enumerate all kinds of unexpected inputs to sanitize. We believe that secure programming should focus on specifying programmers' intentions as opposed to their non-intentions. We introduce a concept called *DOM-tree type*, which expresses the set of DOM trees that an app expects to see during execution, so an exploit will be caught as a type violation. With insights into the HTML standard and the Chromium engine, we build the DOM-tree type mechanism into the Electron platform. The evaluations show that the methodology is practical, and it secures all vulnerable apps that we found in the study.

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